process will have useful lifetimes that are independent with a common unknown distribution F. Physical reasons sometimes suggest the parametric form of the distribution F; for instance, it may lead us to believe that F is a normal distribution, or that F is an exponential distribution. In such cases, we are confronted with a parametrical statistical problem in which we would want to use the observed data to estimate the parameters of F. For instance, if F were assumed to be a normal distribution, then we would want to estimate its mean and variance; if F were assumed to be exponential, we would want to estimate its mean. In other situations, there might not be any physical justification for supposing that F has any particular form; in this case the problem of making inferences about F would constitute a nonparametric inference problem. ■

In this chapter, we will be concerned with the probability distributions of certain statistics that arise from a sample, where a *statistic* is a random variable whose value is determined by the sample data. Two important statistics that we will discuss are the sample mean and the sample variance. In Section 6.2, we consider the sample mean and derive its expectation and variance. We note that when the sample size is at least moderately large, the distribution of the sample mean is approximately normal. This follows from the central limit theorem, one of the most important theoretical results in probability, which is discussed in Section 6.3. In Section 6.4, we introduce the sample variance and determine its expected value. In Section 6.5, we suppose that the population distribution is normal and present the joint distribution of the sample mean and the sample variance. In Section 6.6, we suppose that we are sampling from a finite population of elements and explain what it means for the sample to be a "random sample." When the population size is large in relation to the sample size, we often treat it as if it were of infinite size; this is illustrated and its consequences are discussed.

6.2 The sample mean

Consider a population of elements, each of which has a numerical value attached to it. For instance, the population might consist of the adults of a specified community and the value attached to each adult might be his or her annual income, or height, or age, and so on. We often suppose that the value associated with any member of the population can be regarded as being the value of a random variable having expectation μ and variance σ^2. The quantities μ and σ^2 are called the *population mean* and the *population variance*, respectively. Let $X_1, X_2, \ldots, X_n$ be a sample of values from this population. The sample mean is defined by

$$\overline{X} = \frac{X_1 + \cdots + X_n}{n}$$

Since the value of the sample mean $\overline{X}$ is determined by the values of the random variables in the sample, it follows that $\overline{X}$ is also a random variable. Its

expected value and variance are obtained as follows:

$$\begin{aligned} E[\overline{X}] &= E\left[\frac{X_1 + \cdots + X_n}{n}\right] \\ &= \frac{1}{n}(E[X_1] + \cdots + E[X_n]) \\ &= \mu \end{aligned}$$

and

$$\begin{aligned} \text{Var}(\overline{X}) &= \text{Var}\left(\frac{X_1 + \cdots + X_n}{n}\right) \\ &= \frac{1}{n^2}[\text{Var}(X_1) + \cdots + \text{Var}(X_n)] \quad \text{by independence} \\ &= \frac{n\sigma^2}{n^2} \\ &= \frac{\sigma^2}{n} \end{aligned}$$

where μ and σ^2 are the population mean and variance, respectively. Hence, the expected value of the sample mean is the population mean μ whereas its variance is $1/n$ times the population variance. As a result, we can conclude that $\overline{X}$ is also centered about the population mean μ, but its spread becomes more and more reduced as the sample size increases. Figure 6.1 plots the probability density function of the sample mean from a standard normal population for a variety of sample sizes.

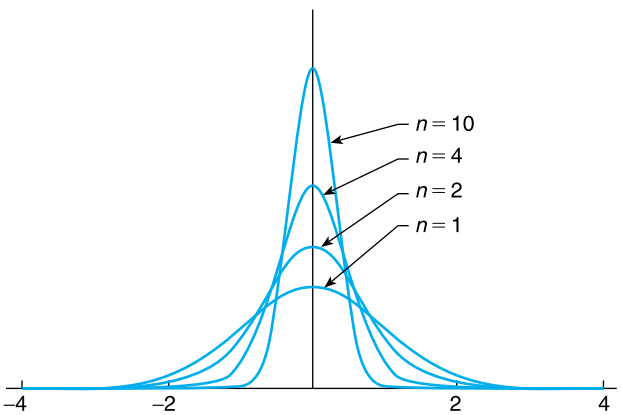


FIGURE 6.1
Densities of sample means from a standard normal population.

6.3 The central limit theorem

In this section, we will consider one of the most remarkable results in probability — namely, the *central limit theorem*. Loosely speaking, this theorem asserts that the sum of a large number of independent random variables has a distribution that is approximately normal. Hence, it not only provides a simple method for computing approximate probabilities for sums of independent random variables, but it also helps explain the remarkable fact that the empirical frequencies of so many natural populations exhibit a bell-shaped (that is, a normal) curve.

In its simplest form, the central limit theorem is as follows:

Theorem 6.3.1 (The central limit theorem). Let $X_1, X_2, \ldots, X_n$ be a sequence of independent and identically distributed random variables each having mean μ and variance σ^2. Then for n large, the distribution of

$$X_1 + \cdots + X_n$$

is approximately normal with mean $n\mu$ and variance $n\sigma^2$.

It follows from the central limit theorem that

$$\frac{X_1 + \cdots + X_n - n\mu}{\sigma\sqrt{n}}$$

is approximately a standard normal random variable; thus, for n large,

$$P\left\{\frac{X_1 + \cdots + X_n - n\mu}{\sigma\sqrt{n}} < x\right\} \approx P\{Z < x\}$$

where Z is a standard normal random variable.

Example 6.3.a. An insurance company has 25,000 automobile policy holders. If the yearly claim of a policy holder is a random variable with mean 320 and standard deviation 540, approximate the probability that the total yearly claim exceeds 8.3 million.

Solution. Let X denote the total yearly claim. Number the policy holders, and let X_i denote the yearly claim of policy holder i. With $n = 25{,}000$, we have from the central limit theorem that $X = \sum_{i=1}^{n} X_i$ will have approximately a normal distribution with mean $320 \times 25{,}000 = 8 \times 10^6$ and standard deviation $540\sqrt{25{,}000} = 8.5381 \times 10^4$. Therefore,

$$P\{X > 8.3 \times 10^6\} = P\left\{\frac{X - 8 \times 10^6}{8.5381 \times 10^4} > \frac{8.3 \times 10^6 - 8 \times 10^6}{8.5381 \times 10^4}\right\}$$

$$= P\left\{\frac{X - 8 \times 10^6}{8.5381 \times 10^4} > \frac{.3 \times 10^6}{8.5381 \times 10^4}\right\}$$

$$\approx P\{Z > 3.51\} \qquad \text{where } Z \text{ is a standard normal}$$
$$\approx .00023$$

Thus, there are only 2.3 chances out of 10,000 that the total yearly claim will exceed 8.3 million.

Of course, we could have used R to calculate the desired probability:

```
> 1 − pnorm(8.3 ∗ (10)^6, 320 ∗ 25000, 540 ∗ sqrt(25000))
[1] 0.0002210042
```

■

Example 6.3.b. Civil engineers believe that W, the amount of weight (in units of 1000 pounds) that a certain span of a bridge can withstand without structural damage resulting, is normally distributed with mean 400 and standard deviation 40. Suppose that the weight (again, in units of 1000 pounds) of a car is a random variable with mean 3 and standard deviation .3. How many cars would have to be on the bridge span for the probability of structural damage to exceed .1?

Solution. Let P_n denote the probability of structural damage when there are n cars on the bridge. That is,

$$\begin{aligned} P_n &= P\{X_1 + \cdots + X_n \geq W\} \\ &= P\{X_1 + \cdots + X_n - W \geq 0\} \end{aligned}$$

where X_i is the weight of the ith car, $i = 1, \ldots, n$. Now it follows from the central limit theorem that $\sum_{i=1}^{n} X_i$ is approximately normal with mean $3n$ and variance $.09n$. Hence, since W is independent of the $X_i, i = 1, \ldots, n$, and is also normal, it follows that $\sum_{i=1}^{n} X_i - W$ is approximately normal, with mean and variance given by

$$E\left[\sum_{1}^{n} X_i - W\right] = 3n - 400$$

$$\text{Var}\left(\sum_{1}^{n} X_i - W\right) = \text{Var}\left(\sum_{1}^{n} X_i\right) + \text{Var}(W) = .09n + 1{,}600$$

Thus,

$$\begin{aligned} P_n &= P\left\{\frac{X_1 + \cdots + X_n - W - (3n - 400)}{\sqrt{.09n + 1{,}600}} \geq \frac{-(3n - 400)}{\sqrt{.09n + 1{,}600}}\right\} \\ &\approx P\left\{Z \geq \frac{400 - 3n}{\sqrt{.09n + 1{,}600}}\right\} \end{aligned}$$

where Z is a standard normal random variable. Now $P\{Z \geq 1.28\} \approx .1$, and so if the number of cars n is such that

$$\frac{400 - 3n}{\sqrt{.09n + 1{,}600}} \leq 1.28$$

or

$$n \geq 117$$

then there is at least 1 chance in 10 that structural damage will occur. ■

One of the most important applications of the central limit theorem is in regard to binomial random variables. Since such a random variable X having parameters (n, p) represents the number of successes in n independent trials when each trial is a success with probability p, we can express it as

$$X = X_1 + \cdots + X_n$$

where

$$X_i = \begin{cases} 1 & \text{if the } i\text{th trial is a success} \\ 0 & \text{otherwise} \end{cases}$$

Because

$$E[X_i] = p, \qquad \text{Var}(X_i) = p(1 - p)$$

it follows from the central limit theorem that for n large

$$\frac{X - np}{\sqrt{np(1 - p)}}$$

will approximately be a standard normal random variable [see Figure 6.2, which graphically illustrates how the probability mass function of a binomial (n, p) random variable becomes more and more "normal" as n becomes larger and larger].

Example 6.3.c. The ideal size of a first-year class at a particular college is 150 students. The college, knowing from past experience that, on the average, only 30 percent of those accepted for admission will actually attend, uses a policy of approving the applications of 450 students. Compute the probability that more than 150 first-year students attend this college.

Solution. Let X denote the number of students that attend; then assuming that each accepted applicant will independently attend, it follows that X is a binomial random variable with parameters $n = 450$ and $p = .3$. Since the binomial is a discrete and the normal a continuous distribution, it is best to compute

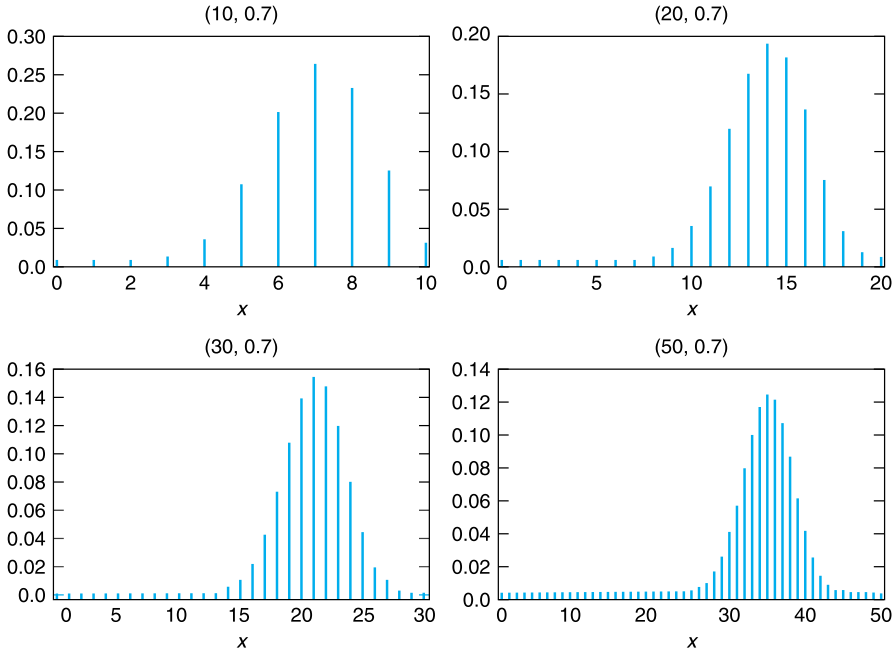


FIGURE 6.2
Binomial probability mass functions converging to the normal density.

$P\{X = i\}$ as $P\{i - .5 < X < i + .5\}$ when applying the normal approximation (this is called the continuity correction). This yields the approximation

$$P\{X > 150.5\} = P\left\{\frac{X - (450)(.3)}{\sqrt{450(.3)(.7)}} \geq \frac{150.5 - (450)(.3)}{\sqrt{450(.3)(.7)}}\right\}$$
$$\approx P\{Z > 1.59\} = .06$$

Hence, only 6 percent of the time do more than 150 of the first 450 accepted actually attend. ■

It should be noted that we now have two possible approximations to binomial probabilities: The Poisson approximation, which yields a good approximation when n is large and p small, and the normal approximation, which can be shown to be quite good when $np(1 - p)$ is large. [The normal approximation will, in general, be quite good for values of n satisfying $np(1 - p) \geq 10$.]

6.3.1 Approximate distribution of the sample mean

Let $X_1, \ldots, X_n$ be a sample from a population having mean μ and variance σ^2. The central limit theorem can be used to approximate the distribution of the

sample mean

$$\overline{X} = \sum_{i=1}^{n} X_i/n$$

Since a constant multiple of a normal random variable is also normal, it follows from the central limit theorem that $\overline{X}$ will be approximately normal when the sample size n is large. Since the sample mean has expected value μ and standard deviation $\sigma/\sqrt{n}$, it then follows that

$$\frac{\overline{X} - \mu}{\sigma/\sqrt{n}}$$

has approximately a standard normal distribution.

Example 6.3.d. The weights of a population of workers have mean 167 and standard deviation 27.

(a) If a sample of 36 workers is chosen, approximate the probability that the sample mean of their weights lies between 163 and 170.
(b) Repeat part (a) when the sample is of size 144.

Solution. Let Z be a standard normal random variable.

(a) It follows from the central limit theorem that $\overline{X}$ is approximately normal with mean 167 and standard deviation $27/\sqrt{36} = 4.5$. Therefore, with Z being a standard normal random variable,

$$\begin{aligned} P\{163 < \overline{X} < 170\} &= P\left\{\frac{163-167}{4.5} < \frac{\overline{X}-167}{4.5} < \frac{170-167}{4.5}\right\} \\ &\approx P\{-.8889 < Z < .8889\} \\ &= P\{Z < .8889\} - P\{Z < -.8889\} \\ &= 2P\{Z < .8889\} - 1 \\ &\approx 2 * \text{ pnorm}(.8889) - 1 \\ &= 0.6259432 \end{aligned}$$

(b) For a sample of size 144, the sample mean will be approximately normal with mean 167 and standard deviation $27/\sqrt{144} = 2.25$. Therefore,

$$\begin{aligned} P\{163 < \overline{X} < 170\} &= P\left\{\frac{163-167}{2.25} < \frac{\overline{X}-167}{2.25} < \frac{170-167}{2.25}\right\} \\ &= P\left\{-1.7778 < \frac{\overline{X}-167}{4.5} < 1.7778\right\} \end{aligned}$$

$$\approx 2P\{Z < 1.7778\} - 1$$
$$= 2 * \text{ pnorm}(1.7778) - 1$$
$$= 0.9245633$$

Thus increasing the sample size from 36 to 144 increases the probability from .6259 to .9246. ■

Example 6.3.e. An astronomer wants to measure the distance from her observatory to a distant star. However, due to atmospheric disturbances, any measurement will not yield the exact distance d. As a result, the astronomer has decided to make a series of measurements and then use their average value as an estimate of the actual distance. If the astronomer believes that the values of the successive measurements are independent random variables with a mean of d light-years and a standard deviation of 2 light-years, how many measurements need she make to be at least 95 percent certain that her estimate is accurate to within $\pm$ .5 light-years?

Solution. If the astronomer makes n measurements, then $\overline{X}$, the sample mean of these measurements, will be approximately a normal random variable with mean d and standard deviation $2/\sqrt{n}$. Thus, the probability that it will lie between $d \pm .5$ is obtained as follows:

$$P\{-.5 < \overline{X} - d < .5\} = P\left\{\frac{-.5}{2/\sqrt{n}} < \frac{\overline{X} - d}{2/\sqrt{n}} < \frac{.5}{2/\sqrt{n}}\right\}$$
$$\approx P\{-\sqrt{n}/4 < Z < \sqrt{n}/4\}$$
$$= 2P\{Z < \sqrt{n}/4\} - 1$$

where Z is a standard normal random variable.

Thus, the astronomer should make n measurements, where n is such that

$$2P\{Z < \sqrt{n}/4\} - 1 \geq .95$$

or, equivalently,

$$P\{Z < \sqrt{n}/4\} \geq .975$$

Since $P\{Z < 1.96\} = .975$, it follows that n should be chosen so that

$$\sqrt{n}/4 \geq 1.96$$

That is, at least 62 observations are necessary. ■